

IRIS FLOWER CLASSIFICATION USING MACHINE LEARNING

Internship Project Report

Submitted By

Name: Saikumar B

Course: Bachelor of Technology (Computer Science Engineering)

Year: Second Year

College : PRESIDENCY UNIVERSITY

Submitted To

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Project Details

Project Title: Supervised Classification Model using Iris Dataset

Algorithms Used:

- **Logistic Regression**
- **Random Forest Classifier**

Tools & Technologies

- **Python**
- **Jupyter Notebook**
- **Scikit-Learn**
- **Pandas**
- **Matplotlib**
- **Seaborn**

Submission Date

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Iris Flower Classification Using Machine Learning

Introduction

The objective of this project is to build and evaluate a supervised machine learning classification model using the Iris dataset. The Iris dataset is one of the most widely used datasets for classification tasks and contains measurements of iris flowers belonging to three different species.

Dataset Description

The Iris dataset consists of 150 samples with four input features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

The dataset contains three target classes:

- Setosa
- Versicolor
- Virginica

Methodology

Data Preprocessing

The dataset was loaded using Scikit-Learn. The data was divided into training and testing sets using an 80:20 ratio. Standardization was applied to improve model performance and ensure consistency across features.

Algorithms Used

Two supervised machine learning algorithms were implemented:

1. Logistic Regression
2. Random Forest Classifier

Model Evaluation

The models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score
- Cross Validation

Results

Logistic Regression

- Accuracy: 100%

```
[6]: from sklearn.metrics import accuracy_score

print("Logistic Regression Accuracy:",
      accuracy_score(y_test, lr_pred))

print("Random Forest Accuracy:",
      accuracy_score(y_test, rf_pred))
```

```
Logistic Regression Accuracy: 1.0
Random Forest Accuracy: 1.0
```

Random Forest Classifier

- Accuracy: 100%

```
Logistic Regression Accuracy: 1.0
Random Forest Accuracy: 1.0
```

Classification Metrics

- Precision: 1.00
- Recall: 1.00
- F1-Score: 1.00

```
from sklearn.metrics import classification_report

print(classification_report(y_test, rf_pred))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Cross Validation

- Average Cross Validation Score: 96.67%

```
[9]: from sklearn.model_selection import cross_val_score
```

```
scores = cross_val_score(rf, X, y, cv=5)
```

```
print("Cross Validation Scores:", scores)
```

```
print("Average CV Score:", scores.mean())
```

```
Cross Validation Scores: [0.96666667 0.96666667 0.93333333 0.96666667 1.
Average CV Score: 0.9666666666666668]
```

ROC-AUC Score

- ROC-AUC Score: 1.00

```
[10]: from sklearn.metrics import roc_auc_score
```

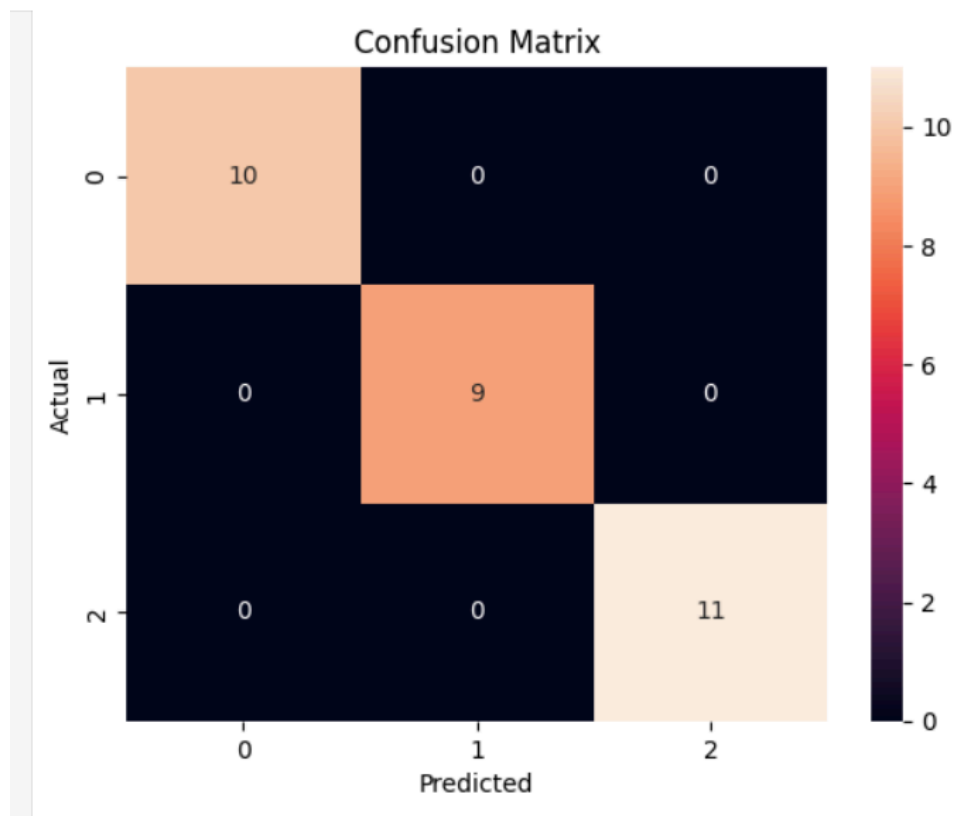
```
rf_probs = rf.predict_proba(X_test)
```

```
roc_auc = roc_auc_score(
    y_test,
    rf_probs,
    multi_class='ovr'
)
```

```
print("ROC-AUC Score:", roc_auc)
```

```
ROC-AUC Score: 1.0
```

Confusion Matrix Analysis



The confusion matrix shows that all test samples were classified correctly. No misclassifications were observed, indicating excellent model performance on the Iris dataset.

Conclusion

The project successfully implemented and evaluated supervised classification models using the Iris dataset. Both Logistic Regression and Random Forest achieved perfect accuracy on the test dataset. Cross-validation results also demonstrated strong generalization capability. The Random Forest model performed exceptionally well and produced accurate predictions for all classes.

Tools and Technologies

- Python
- Jupyter Notebook
- Scikit-Learn
- Pandas
- Matplotlib
- Seaborn

